

GO HPC docs website - homework

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20260430

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Topics

\1. Abbreviations, glossary and acronyms

\2. “Homework”: which docs are needed?

Some external experts give advice on documentation:

- Bran van der Meer
- Brian P. Hogan

Results of the homework:

- EW
- EL
- GO

\3. Use of url's in otrs tickets

\4. New in zensical

- lightbox demo
- <https://zensical.org/docs/authoring/images/?h=light#lightbox-and-zoom>

```
[project.markdown_extensions.attr_list]
[project.markdown_extensions.md_in_html]
[project.markdown_extensions.pydownx.blocks.caption]
[project.markdown_extensions.zensical.extensions.glightbox]
```

Abbreviations, Glossary and Acronyms

- zensical (0.0.43) has support for abbreviations - inline (defined in one document) - by using a central file
e.g. includes/abbreviations
 - see documentation
 - see sample

Glossary

an alphabetical list of terms in a particular domain of knowledge with the definitions for those terms

Why?

- HPC has a lot of specialist words
- Most software on docs.hpc has no description

*[HPC]: High Performance Computing

How?

- part of the glossary can be extracted from the descriptions in
 - EasyBuild
 - EESSI
- other possible sources:
 - man page
 - apt
 - debian screenshots
 - wikipedia
 - (git) repo
 - website of (commercial) producer of the software
 - dictionaries such as
 - * Free On-line Dictionary of Computing
 - * V.E.R.A. - Virtual Entity of Relevant Acronyms
- Which words?
 - how can we select the needs we need?
 - how many (estimation, based easybuild: 6K+)
 - how to keep up-to-date
 - zensical adds the extra html code based on the include with a lot of glossary the build might take longer
- Where?
 - The glossary can be one separate page which can be consulted
 - or a separate file per topic `**about__topic__.md**`
 - as a complement to the script generated `available_software/detail/topic.md` file
 - both files should link to each other
 - this page(s) will give results via the search option
 - when used as an include, the `<abbr>` tag will be included and the title will be shown as a pop-up (see tool-tip)
`<abbr title="High Performance Computing">HPC</abbr>`

Abbreviations and Acronyms

E.g. HPC = High Performance Computing

- can be build from glossary
- should be short - preferably less than 80 chars.
- longer info can be used via instant or automatic preview
 - included in zensical
 - some browsers have also this functionality (screenshot)

Example APOST3D aka APOST-3D

Source:

<https://github.com/easybuilders/easybuild-easyconfigs/blob/develop/easybuild-easyconfigs/a/APOST3D/APOST3D-20240527-intel-compilers-2023.1.0.eb>

Glossary term

*[APOST3D]: Open-source APOST-3D software features a large number of wavefunction analysis tools.

Abbreviation

*[APOST3D]: wavefunction analysis tools connecting classical chemical concepts with the electronic structure of molecules.

Acronym

*[APOST3D]: This program performs *a posteriori*

A) *Bond order and valence analysis and*

B) *"Chemical Energy Component Analysis" (CECA)*

of the *ab initio* SCF wave functions (RHF or UHF) produced by the widely used "Gaussian" system (G92, G94, G98).

<http://occam.ttk.hu/programs/readme.html>

Not really an acronym

Conversion of an easybuild eb file

The most useful parts are

- **name**, e.g. APOST3D
- **homepage**, e.g. <https://github.com/mgimferrer/APOST3D>
- **description**, e.g. > Open-source APOST-3D software features a large number of wavefunction > analysis tools developed over the past 20 years, aiming at connecting > classical chemical concepts with the electronic structure of molecules. > APOST-3D relies on the identification of the atom in the molecule (AIM), and > several analysis tools are implemented in the most general way so that they > can be used in combination with any chosen AIM. A Fortran-based code > developed at the Universitat de Girona (UdG) by P. Salvador and > collaborators.

[APOST3D]: wavefunction analysis tools connecting classical chemical concepts with the electronic structure of molecules. [APOST-3D]: wavefunction analysis tools connecting classical chemical concepts with the electronic structure of molecules.

Bran van der Meer

Five Principles for Good Documentation - YouTube <https://www.youtube.com/watch?v=H0GKczmzYzI>

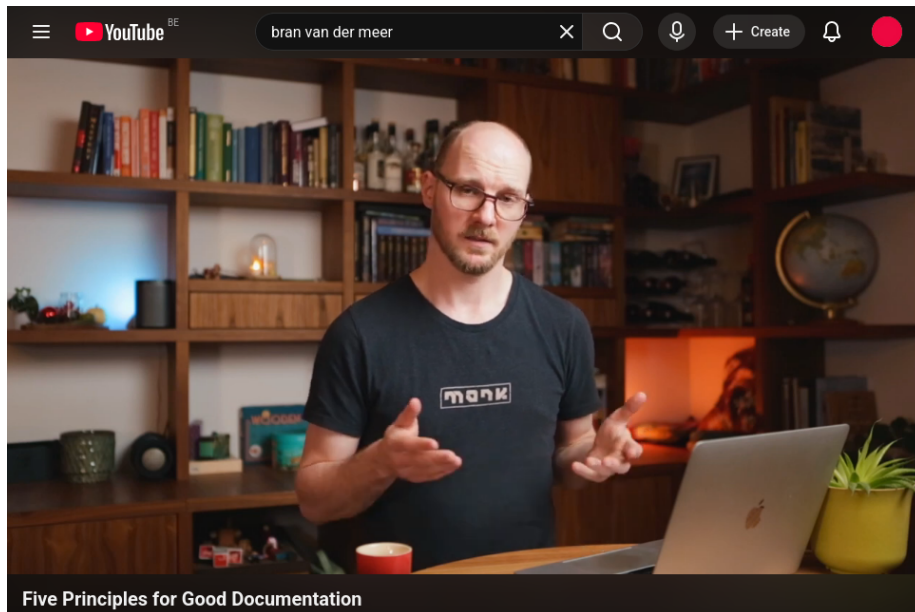


Figure 1: youtube

Five principles

1. relevant
2. timeless
3. tested
4. centralised, accessible
5. collaborative

Sections

- overview
- domain
- architecture
 - example of hpc architecture i.e. building blocks
 - <https://neysa.ai/blog/hpc-architecture/>
 - <https://hpc101.readthedocs.io/en/latest/tutorial/session1/arch.html>
- getting started
- code conventions
 - rules, design patterns, naming
 - why are they so
 - not the things you can automate with linter and pre-commit
- key dependencies (if needed)
- faq / troubleshooting
 - build as faq's become clear not by thinking what would be a potential faq

Five common problems

1. hard to understand
2. incomplete or non existent
3. out of date
4. overly specific
5. solving the wrong problem

Brian P. Hogan

homepage

Audience for our docs

- People who know the feature they need but **can't remember** how to use it.
- People who have a problem but **don't know how to use your product** to solve it.
- People who have a problem but **don't know you have a solution** for it.
- People who **don't know they have the problem you solve**.

See newsletter 53

Which texts the user really needs (and why) (EW)

I tried to put myself in the shoes of a **novice user with no prior knowledge of the domain** who wants to run tasks on the high performance computing (HPC) infrastructure.

From that perspective, these topics are important:

1. How to connect to the HPC via Open On Demand (OOD)?
2. How to launch an interactive task on the debug cluster? E.g. a Jupyter notebook (via the pull-down menu)
3. Where to store my files? How much storage room is available?
4. How to submit a job on another cluster (with a queue)?
5. How to request more storage (join Virtual Organisation)?

Which texts we really need (and why)

TOC

- About the VSC Accountpage (AP)
- Requesting a Vlaams supercomputer Centrum (VSC) account
- Policies
- Web portal
- How to ask for support
- Overview of clusters
- Quota
- SSH page

About the VSC Accountpage (AP)

This about https://docs.vscentrum.be/accounts/vsc_account.html

Short description of what **the VSC accountpage** is and what users can do there:

- request a VSC account (link to ‘Request a VSC account’ page - can be the one of vscentrum https://docs.vscentrum.be/accounts/vsc_account.html#new-vsc-account but watch out, the UGent tab there refers to <https://www.ugent.be/hpc/en/access>)
- monitor their quota (link to ‘Quota’ page)
- request group/Virtual Organisation (VO) membership (link to ‘VO’ page or <https://docs.vscentrum.be/accounts/vo.html>)
- request new VO (link to ‘VO’ page or <https://docs.vscentrum.be/accounts/vo.html>)
- add additional SSH keys (link to ‘SSH’ page)

Mention 1-hour delay for actions on AP to become active on the system

Why - because support often gets requests for things users can do themselves

Requesting a Vlaams supercomputer Centrum (VSC) account

- Steps to request a VSC account
- By default should be **without SSH key**.
- Option to create an SSH key (link to ‘SSH’ page)
- Link to ‘Policies’ page for who can get an account

Partly covered in https://docs.vscentrum.be/accounts/vsc_account.html#applying-for-your-vsc-account (but watch out, the UGent tab there refers to <https://www.ugent.be/hpc/en/access>)

Why - first step to get started with Tier-2 UGent is to get a VSC account

Policies

- Overview of who can get a VSC account can we link to this page: <https://www.ugent.be/hpc/en/access/policy/access?>
- Acknowledgement for publication when users use HPC-UGent
- Fairshare policy for job scheduling can link to https://slurm.schedmd.com/fair_tree.html for more details (or https://slurm.schedmd.com/SLUG19/Priority_and_Fair_Trees.pdf?)

Why - support often gets questions on who can get access and how

Web portal

Page on how to connect to and use the web portal:

- potential need for firewall to connect (maybe link to docs.vscentrum https://docs.vscentrum.be/security_measures_20200520.html)
- file browser for upload/download
- jobs overview
- shell access to cluster

- interactive apps
- submit support ticket from portal

Why - we're pushing more people to use the web portal so there must be a page to get them started (especially beginners)

How to ask for support

Page with list of things to provide when contacting us + hpc@ugent.be

Content below taken from one of our prefab answers in OTRS:

Please help us help you by providing more information in a reply to this message (if it is applicable to your question, or not yet provided):

- Your VSC account ID
- Which software application(s) you are using
(which modules you are loading, etc.)
- A short description of what you want to achieve
- Name of cluster(s) being used
- Relevant job ID(s)
- The command used to submit the job(s)
- All error and/or warning messages that may be relevant
- Location of the job script(s) you are using in your VSC account
- Location of the job output files in your VSC account (both regular output + errors/warnings)

Link to this page from bottom of every other page

Why - if more users would provide relevant info in their first message it would save support a lot of time

Overview of clusters

- Overview of clusters we currently have
 - their specs
 - subdivision into CPU, GPU and debug
- Info on user limits for debug cluster
- What is our default cluster

Why - people need to know which clusters we have to choose which one fits their needs

Quota

- Describe the personal quota per user
- in \$VSC_HOME, \$VSC_SCRATCH and \$VSC_DATA
- Disclaimer that there is NO BACKUP
- link to 'UGent shares' page to give them a way to back up

- Link to ‘VSC Accountpage’ for consulting their usage
- What if they run out of storage: clean up or join a VO (link to ‘VO’ page)
- What if they get message ‘Diskspace full’ but quota not reached: inode limit
- Useful commands how to check for big (hidden) files

Why - support gets lots of questions on storage issues

UGent shares

- Info on how to access UGent shares on our systems: globus/kinit
- Disclaimer that this is only for staging and only on login nodes, not for live computations

Why - very UGent specific and users need instructions to do it right.

SSH

- Start with some kind of disclaimer that default access is via web portal (link to ‘Web portal’ page)
- If they want SSH they’ll need to have some basic linux knowledge and we can’t hold their hand on that.
- Can link to https://docs.vscentrum.be/accounts/generating_keys.html#cryptographic-keys it has examples on how to create keys in windows, linux and mac.
 - Link to ‘WSL’ page for windows users that don’t want to use powershell.
- Mention how to upload new key to AP (link to page ‘VSC Accountpage’)
- How to connect via ssh to our systems
- Fingerprints of login.hpc.ugent.be for users to verify
- generated by us, on the website
- command to verify before making a connection

```
ssh-keyscan login.hpc.ugent.be | ssh-keygen -lf -
```

Why - for users that want shell access via SSH, we should try to cover only our UGent-specific stuff

Windows Subsystem for Linux (WSL)

- **Very short** description of what WSL is + link to external resources how to set it up
- Disclaimer that we don’t provide support on this
- Link to ‘SSH’ page for how to create ssh keypair
- Tip for combining it with UGent VPN:
 - needs virtioproxy mode and
 - must start VPN first, then WSL2

Why - for windows users that can't/won't use powershell, the UGent VPN thing is again very UGent specific

Software availability

- Describe module concept
- How to check if specific software is available:
 - overview of module commands `module avail`, `module show`, `module spider`, ...
- Indicate that not all software is available on all clusters
(link to 'Overview of clusters' page)
- How to request more software
 - install request form (+ link) or
 - install it yourself (link to 'Installing software yourself' page)
- Combining multiple modules: (link to 'Software compatibility' page)

Why - users have to be able to check which software is available otherwise they won't use our systems

Software compatibility

- Link to 'Software availability' page for basic info
- Explain toolchain version concept briefly (or link to external source if there is a good one)
- If combining multiple modules they need same toolchain version
- Example on how to test in a shell session on login node on target cluster by doing `'module load a'` and `'module load b'`
- Give table of which GCCcore is compatible with which foss for example
- What if there is no possible combination of software you want? request software install or DIY

Why - combining software is not always trivial, it deserves some explanation

Installing software yourself

- Warn to not use `$VSC_HOME` for installing because of quota issues
(link to 'Quota' page)
- Python: Warn to use python venvs instead of conda (link to 'Python virtual environments' page)
- R: describe how and where custom user installs end up, that they may need to repeat this per cluster
- Mention EasyBuild can be used by individual users to install in VO (link to 'VO' page)

Why - if we raise the bar to request a software install, we must provide users with info on how to do it themselves (if they want to)

Python virtual environments

Section on how to use the module vsc-venv for creating virtual environments

Why - same as above for “Installing software yourself”

Submitting jobs

- Step-by-step how to submit: swap to right cluster, qsub job script, qstat for status, default name & location of err/output files
- Overview of job script commands (qsub, qstat, qdel, ...) that can be used to check job status in a terminal
- Link to page ‘Job script’ for info on job scripts
- Add link to page ‘Web portal’ section ‘Shell access to cluster’ so we can point web portal users to how to get a terminal

Why - even though focus is on portal, **slurm** is still the motor behind all jobs

Job scripts

- Explain #PBS flags (Portable Batch System)
- Explain basic parts
 - loading modules,
 - copying data,
 - running program,
 - copying data back
- Example of a minimal job script for single-core and multi-core

Why - same reason as above ‘Submitting jobs’

Teaching and training

- Overview of what we offer for teaching and training
- List of info they need to provide to us

Why - used quite a bit by teachers right now and they usually do copy paste the list of info we need into the email

Virtual Organisations (VO)

- Actually I don’t know if we need a separate page on that because it’s pretty well described in <https://docs.vscentrum.be/accounts/vo.html>
- Nevertheless we do often refer users to a VO because either they need more storage, or they are waiting in the queue because someone in the default gent VO ate up all the **fairshare**

FAQ/troubleshooting

The issues I see most often:

- how can I get more storage?
- home directory full (hidden files)?
- my job is still in the queue after x hours why is it not starting?
- can I get more than 72h walltime?

- OOD: why did my job start and immediately complete?

Why - makes it easy for support to point users to the right page with just the info they need

Which texts the user really needs (and why) (GB)

Which documents should be on the site?

(homework April 22, 2026)

- about the hpc infrastructure, what is available, performance data
- why and when (in which case) use hpc
- how to use: conditions to use, see also vsc
- how to log in: portal (OoD), ssh, vpn
- hpc specific terminology
- linux basics
 - <https://dodona.be/nl/courses/4165/series/46395/activities/2055766930/>
 - <https://www.w3schools.com/bash/>
 - <https://labex.io/linuxjourney>
 - <https://linuxsurvival.com>
 - Introduction to the Bash Command Line | Programming Historian
<https://programminghistorian.org/en/lessons/intro-to-bash>
 - getting help
 - * help (!)
 - * man
 - * man pages online
 - <https://man.he.net/>
 - <https://man.freebsd.org/cgi/man.cgi>
 - commands with warnings
 - * rm
 - * -r
- git
 - https://developer.mozilla.org/en-US/docs/Learn_web_development/Core/Version_control
 - <https://swcarpentry.github.io/git-novice/>
- getting data in and out
- running programs with a graphical interface
- running jobs
- which software is available
- what software for what kind of problem
- how to run a specific version of a software program
- (past) training / workshop material
- video's
- security issues
- how to get help
 - Using hpc is not complex but you have to understand what is going on. The best way to start is to ask a friend to show you the way.
 - how does the hpc helpdesk operate, see <https://www.maths.cam.ac.uk/computing/computing-and-it>

- * opening hours, response time
- * what to expect
- how and where to find info on internet
- how to ask for help to our helpdesk
- how to make a **useful** screenshot
- how to describe your problem
- what to include?
- kan je eens proberen om die pagina te bezoeken in een “incognito” browser venster, of eventueel een andere internet browser (Firefox, Chrome, ...)
- dan je gewoon bent om te gebruiken?

-
- privacy
 - terms of service »»»» c33c020 (add ticketlist.ods)

Analysis of OTRS tickets

Maybe we can find the docs we use most for helpdesk tickets?

What url’s are in HPC tickets?

- a lot of noise from mail footers
(are those links useful in otrs?)
- not that many links to docs.hpc
- no access to database, ergo: some kind of scraping to collect the data method

Per domain

2	1	arts.kuleuven.be
2	1	bsky.app
2	1	cran.r-project.org
2	1	docs.google.com
2	1	docs.vscentrum.be
2	1	doi.org
2	1	forms.office.com
2	1	github.ugent.be
2	1	pisa.ugent.be
2	1	textiles.ugent.be
2	1	wiki.csi.cuny.edu
2	2	bioinformaticsworkbook.org
2	2	lcp.elis.ugent.be
2	2	vibtrainingandconferences.be
2	2	vscentrum.be
2	3	event.ugent.be
2	3	febresearchday2025.ugent.be
2	3	research.ugent.be
2	3	twitter.com
2	3	users.ugent.be

2	3	wiki.ugent.be
2	4	login.hpc.ugent.be
2	5	filesender.belnet.be
2	5	go.microsoft.com
2	6	apivsc.ugent.be
2	6	soleway.ugent.be
2	7	account.vscenrum.be
2	7	uc.ugent.be
2	8	eur01.safelinks.protection.outlook.com
2	11	github.com
2	13	helpdesk.ugent.be
2	33	eur03.safelinks.protection.outlook.com
2	38	tiktok.com
2	39	youtube.com
2	42	facebook.com
2	42	instagram.com
2	42	linkedin.com
2	45	pep.pxl.be
2	57	pxl.be
2	125	docs.hpc.ugent.be
2	534	ugent.be

Tickets processed: 28

docs.hpc

0	1	https://docs.hpc.ugent.be/account/#adding-multiple-ssh-public-keys-optional
0	1	https://docs.hpc.ugent.be/gpu
0	1	https://docs.hpc.ugent.be/infrastructure/
0	1	https://docs.hpc.ugent.be/interactive_debug/#shared-gpus
0	1	https://docs.hpc.ugent.be/jupyter/
0	1	https://docs.hpc.ugent.be/jupyter/#using-extra-python-packages
0	1	https://docs.hpc.ugent.be/macOS/available_software/
0	1	https://docs.hpc.ugent.be/running_batch_jobs/#modules
0	1	https://docs.hpc.ugent.be/running_batch_jobs/#specifying-the-cluster-on-which
0	1	https://docs.hpc.ugent.be/sites/hpc_policies/
0	2	https://docs.hpc.ugent.be/troubleshooting/#module-conflicts
0	2	https://docs.hpc.ugent.be/web_portal/
0	3	https://docs.hpc.ugent.be/web_portal/#using-the-hpc-ugent-web-portal
0	5	https://docs.hpc.ugent.be/FAQ/#my-home-directory-is-almost-full-and-i-dont-know
0	7	https://docs.hpc.ugent.be/macOS/
0	22	https://docs.hpc.ugent.be/
0	74	https://docs.hpc.ugent.be/software_installation_requests

Step-by-step guide to using the Whisper transcription on the HPC

Below is a step-by-step guide to using Whisper transcription on the HPC.

Please do not hesitate to send any queries regarding this to <mailto:hpc@ugent.be>

(this address is monitored by several of us)

Connecting

1. In a web browser, navigate to <https://login.hpc.ugent.be>
 - Initially this will show you
“text You need to connect to the firewall app in new tab and wait up to 30s.
Keep the tab open while you are connected.
“text
 - Connect to the VSC firewall at <https://firewall.hpc.kuleuven.be/>
2. Select “Ghent University” and then log in using your Ghent University account details (in the background, this will be linked to your vsc12345¹ HPC account)

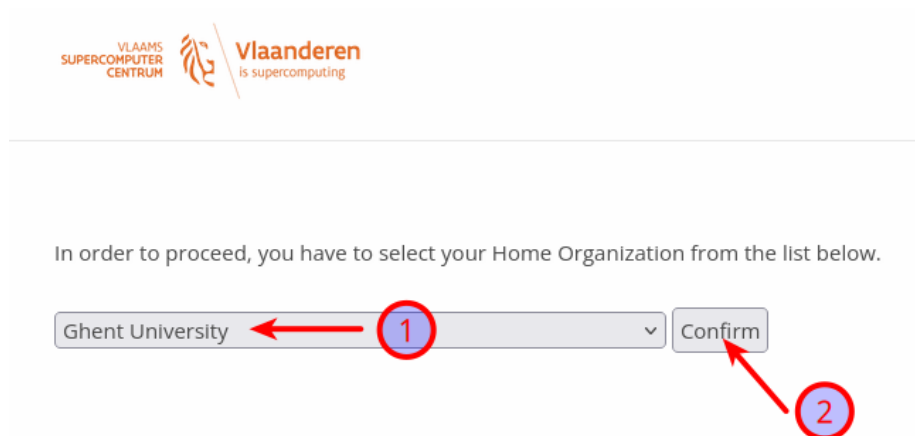


Figure 2: Select

You will see a consent screen if you did not connect to VSC before.

Select the “Yes, continue” button.

Approve the use of the hpc-firewall, select “Authorise”.

Information on your connection to the HPC firewall will be shown.

Important: leave this tab open in your browser.

Open <https://login.hpc.ugent.be> again **in another tab**.

FIXME: another confirmation is needed? confirm, add screenshot

3. You should now see a website with a blue HPC-UGent menu bar at the top.

¹Your vsc account will have other numbers, please adjust.

UNIVERSITEIT GENT

WELKOM
External Accounts Management

In het Nederlands Help

Home > Welkom

https://account.vscentrum.be requires that the information below is transferred.

☒ Remember [Yes, continue](#) [No, cancel](#)

Information that will be sent to https://account.vscentrum.be

User information

Given name

Email

Affiliation

Affiliation at home organization

Person's principal name at home organization

Persistent pseudonymous ID [... Show content](#)

Disclaimer Cookies Toegankelijkheid v. 1.0 © 2021 Universiteit Gent

Figure 3: Consent

Authorize hpc-firewall?

Application requires the following permissions

- Read scope

[Cancel](#) [Authorize](#)

Figure 4: Approve hpc-firewall

HPC Firewall

IP Whitelist | Info

Logged on successfully as vsc .

your vsc account

your ip address

 is granted access since 2026-05-20 11:09:59 [valid until 11:16:29] (will be refreshed).

Figure 5: HPC firewall info

HPC Ugent
Files
Jobs
Clusters
Interactive Apps



OnDemand provides an integrated, single access point for all of your HPC resources.

Status

STEVIN HPC-UGent infrastructure status on Sun, 31 May 2026 10:40:01

cluster	full nodes	free nodes	partial free nodes	total nodes	running jobs	queued jobs	jobs
joltik	3	3	3	9	11	2	
doduo	60	1	30	95	171	6249	
accelgor	1	1	7	9	13	62	
donphan	0	6	9	16	7	6	
gallade	4	0	8	16	133	2176	
shinx	18	0	30	48	130	3346	
litleo	8	0	0	8	16	1072	

Documentation is available at: <https://www.ugent.be/hpc/en/support/documentation.htm>

For a full view of the current loads and queues see: <https://hpc.ugent.be/clusterstate/>

Updates on current system status and maintenance can be found on: <https://www.ugent.be/hpc/en/infrastructure/status>

To contact the support team, send an email to hpc@ugent.be.

****Please be cautious with which extensions you use/install with VS Code, as more and more malware extensions are reported. See: <https://code.visualstudio.com/docs/configure/extensions/extension-runtime-security>**

[Fri 22 May 2026, 09:00 CEST] We have successfully completed the planned maintenance and have mitigated the recent security vulnerability on HPC-UGent Tier-2.

[Thu 28 May 2026, 15:50 CEST] HPC-UGent Tier-2 login nodes were rebooted to disable the CIFS kernel module, to mitigate a known security issue. As a result, access to Ugent shares is temporarily unavailable.**

powered by

OnDemand version: 3.1.16

Figure 6: HPC Dashboard

Uploading files

4. In the blue bar, click on “Files” and then \$VSC_DATA FIX: add screenshot
5. If necessary, create a new folder here (using the “New Directory” option at the top)
6. You can easily upload a file from File Explorer to the website using drag-and-drop:
 - Simply drag the file onto the HPC web browser page
 - Click the green ‘Upload 1 file’ button. Fix: add screenshot

Starting Whisper

7. In the blue bar at the top,
 - click on ‘Interactive Apps’ and
 - then ‘Transcribe’ (bottom menu option) Fix: add screenshot
8. Select the input file you uploaded in steps 4–6:
 - Click the ‘Select Path’ button
 - In the favourites on the left, select the folder ‘/data/gent/123/vsc12345’²
 - If necessary, navigate in the right-hand column to the separate folder you created in step 5 FIX: add link as described in <https://stackoverflow.com/questions/2822089/how-to-link-to-part-of-the-same-document-in-markdown?noredirect=1&qlq=1>
 - Highlight the file you uploaded in blue
 - Then click on “Select Path” at the bottom
9. Select “Dutch” as the “Whisper language” (or another language if the conversation was not conducted in Dutch)
10. Select the “Show advanced options” button
 - Set “Translation target languages” to “Dutch”
 - Under “Advanced Whisper Flavour”, select “WhisperX”
 - Select the “WhisperX speaker diarisation” button
 - Further down, under “Time (hours)”, select ‘6 hours (1/4 day)’
11. Click on “Launch” at the very bottom
 - Your transcription with diarisation will now start.
 - You will receive emails when the calculations begin and when they are complete.

Viewing the transcription with diarisation

12. The final email contains a link that takes you directly to the files (under “Result”)
 - The .txt file contains the text with [SPEAKER_00,01,02,...] labels
 - The .vtt file also contains time codes

Colophon

- this page is based on an OTRS ticket on this topic, compare to <https://docs.hpc.ugent.be/transcribe/>

²Your vsc account will have other numbers, please adjust.

- the screenshots can be shown in a lightbox overlay, available as an zensical extension
 - see the zensical.toml of this site
 - see also <https://www.nngroup.com/articles/overuse-of-overlays/>
- to continue the numbering of a list in markdown see <https://stackoverflow.com/a/18089124/906489> You need to indent with 4 spaces.